Class 12

Fermionic mode optimization

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Fermions with 2-body interactions – generic model

$$H = \sum_{ij} T_{ij} c_i^\dagger c_j + \sum_{ijkl} V_{ijkl} c_i^\dagger c_j^\dagger c_k c_l$$

Initial Hartree-Fock / DFT simulation ➡

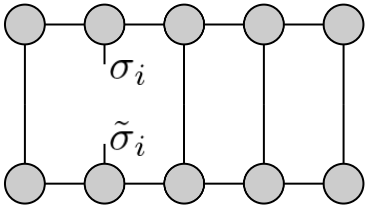
- Active orbitals: $c_i^\dagger \quad i \in 1 \ldots L$
- Matrix elements: $T_{ij}\ ,\ V_{ijkl}$

Let's use DMRG for the ground state! The ground state is probably strongly correlated!

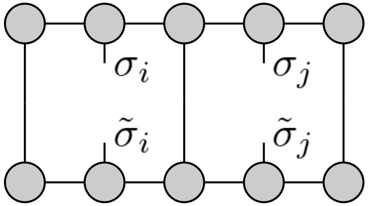
Questions:

- Is the HF-based active space selection good? ➡ **CAS-SCF**
- If it is good, is the single particle basis optimal for DMRG? ➡ **Entanglement based mode optimization**
- What is the optimal ordering of orbitals for MPS truncation? ➡ **Mutual info. based optimal ordering**

Ordering Optimization



$$\mathrm{Tr}_{\mathrm{all}\setminus i}|\Psi\rangle\langle\Psi| = \sum_{\sigma_i\tilde{\sigma}_i} \rho^{(1)}_{\sigma_i\tilde{\sigma}_i}|\sigma_i\rangle\langle\tilde{\sigma}_i|$$

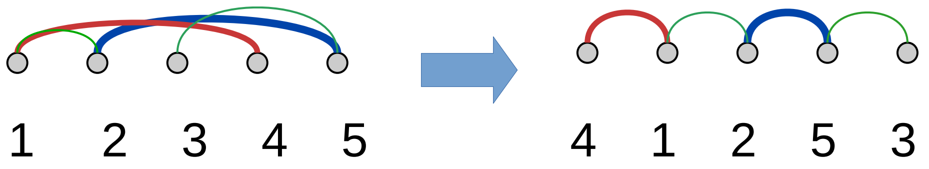


$$\mathrm{Tr}_{\mathrm{all}\setminus\{i,j\}}|\Psi\rangle\langle\Psi| = \sum_{\sigma_i\tilde{\sigma}_i}\sum_{\sigma_j\tilde{\sigma}_j} \rho^{(2)}_{\sigma_i\sigma_j,\tilde{\sigma}_i\tilde{\sigma}_j}|\sigma_i\sigma_j\rangle\langle\tilde{\sigma}_i\tilde{\sigma}_j|$$

1-site and 2-site entropies: $S_i = -\mathrm{tr}\rho^{(1)}_i \log \rho^{(1)}_i$ $\quad S_{ij} = -\mathrm{tr}\rho^{(2)}_{ij} \log \rho^{(2)}_{ij}$

2-site mutual information: $I_{ij} = S_i + S_j - S_{ij}$

Qualitative picture of optimization:



Ordering Optimization (Barcza, Legeza, Marti, Reiher 2011)

$$I_{\text{cost}}^{(\eta)} = \sum_{i,j} I_{ij}(i-j)^{\eta}$$

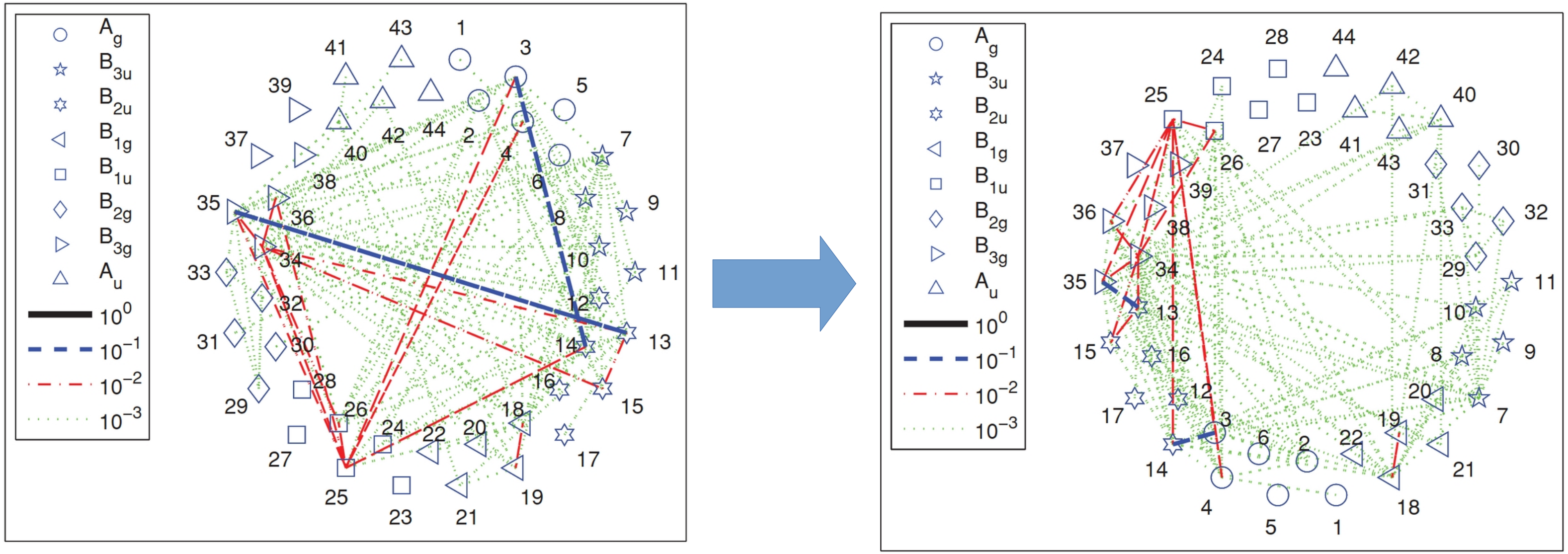
Quasi-optimal solution is easy to find for $\eta = 2$

$$\min_{\{x_i\}} \sum_{ij} I_{ij}(x_i - x_j)^2$$

With constraints: $\sum_i x_i = 0$ & $\sum_i x_i^2 = 1$

Fiedler vector: Second eigenvector of $F_{ij} = \delta_{ij} \sum_k I_{ik} - I_{ij}$

Ordering the entries of the Fiedler vector provides a good ordering



The role of the 1-particle basis

Tight-binding model (spinless for simplicity):

$$H = -t \sum_x \left(c_x^\dagger c_{x+1} + c_{x+1}^\dagger c_x \right)$$

Half filling: $n_x = \langle c_x^\dagger c_x \rangle = 0.5$

1-site entropy is maximal!

$$S_x = -n_x \log n_x - (1 - n_x) \log(1 - n_x) = \log 2$$

Highly entangled state!

Transform to momentum space (Fourier basis)

$$d_k^\dagger = \sum_x \frac{e^{-ikx}}{\sqrt{L}} c_x^\dagger \qquad H = \sum_{k \in [-\pi, \pi]} -2t \cos(k) d_k^\dagger d_k$$

$$\langle d_k^\dagger d_k \rangle = \Theta(\cos(k))$$

1 for occupied, 0 for empty orbitals

This is a product state (single Slater determinant!)

Fermionic mode transformation for interacting models

$$H = \sum_{ij} T_{ij} c_i^\dagger c_j + \sum_{ijkl} V_{ijkl} c_i^\dagger c_j^\dagger c_k c_l$$

$$d_i^\dagger = \sum_j U_{ij} c_j^\dagger$$

1-particle (L x L) unitary

$$H = \sum_{ij} \tilde{T}_{ij} d_i^\dagger d_j + \sum_{ijkl} \tilde{V}_{ijkl} d_i^\dagger d_j^\dagger d_k d_l$$

$$\tilde{T}_{ij} = \sum_{kl} U_{ki}^* T_{kl} U_{lj}$$

$$\tilde{V}_{ijkl} = \sum_{mnop} U_{mi}^* U_{nj}^* V_{mnop} U_{ok} U_{pl}$$

Transformation of the (approximate) state vector:

$$\underline{\underline{U}} = e^{i\underline{\underline{W}}} \cong \left(1 + \frac{i}{N}\underline{\underline{W}}\right)^N$$

$$G(U) = \lim_{N\to\infty} \left(1 + \frac{i}{N}\sum_{ij} W_{ij} c_i^\dagger c_j\right)^N$$

$$|\tilde{\Psi}\rangle = G(U)|\Psi\rangle$$

Fermionic mode transformation for two sites (real case)

$$\underline{\underline{U}}_\varphi = \begin{pmatrix} \cos\varphi & \sin\varphi \\ -\sin\varphi & \cos\varphi \end{pmatrix}$$

2-site basis: $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$

These are invariant

These states transform

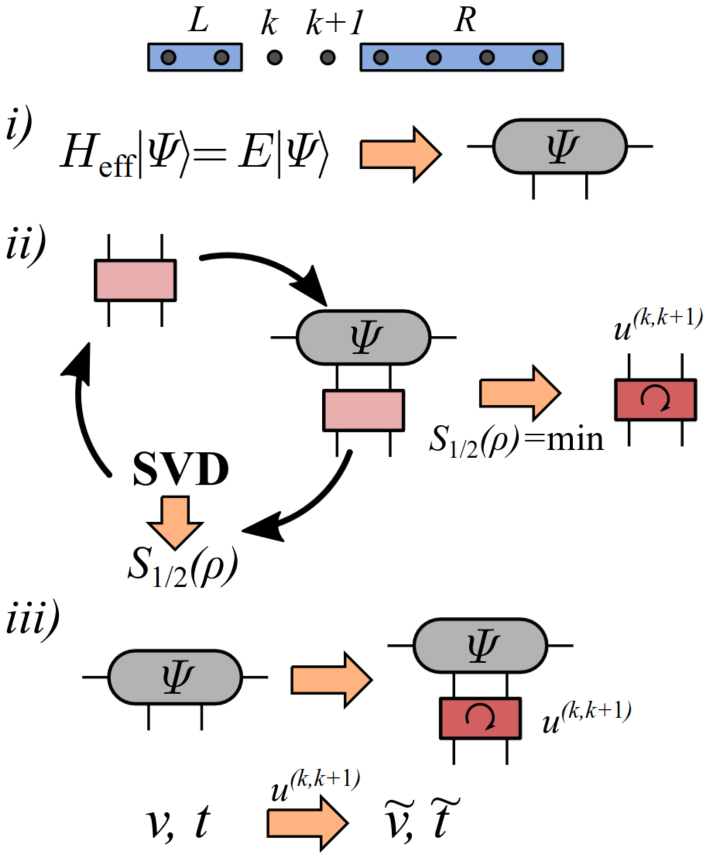
$$G(U_\varphi) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\varphi & \sin\varphi & 0 \\ 0 & -\sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

This is a two-site unitary gate!

DMRG with Local mode optimization

Motivation:

- Every unitary can be factorized as product of 2-site unitaries
- Any permutation can be reached by nearest-neighbor swaps



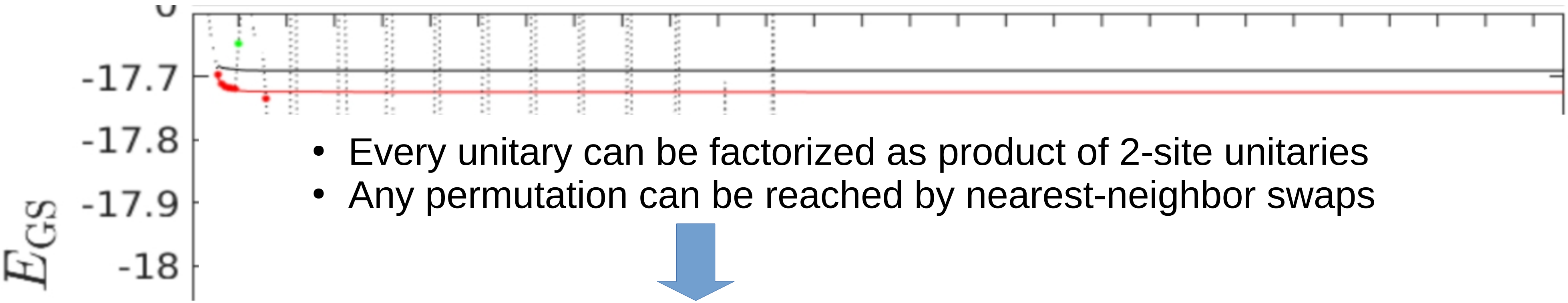
Local optimization step for a nearest-neighbor 2-site unitary

- The total block entropy is minimized to reduce the required bond dimension.
- Rényi-1/2 instead of von Neumann

$$S_{1/2}(\rho) = 2\log \mathrm{Tr}(\sqrt{\rho})$$

Larger contribution from small Schmidt weights: optimization attemts to remove these

Local mode optimization gets stuck

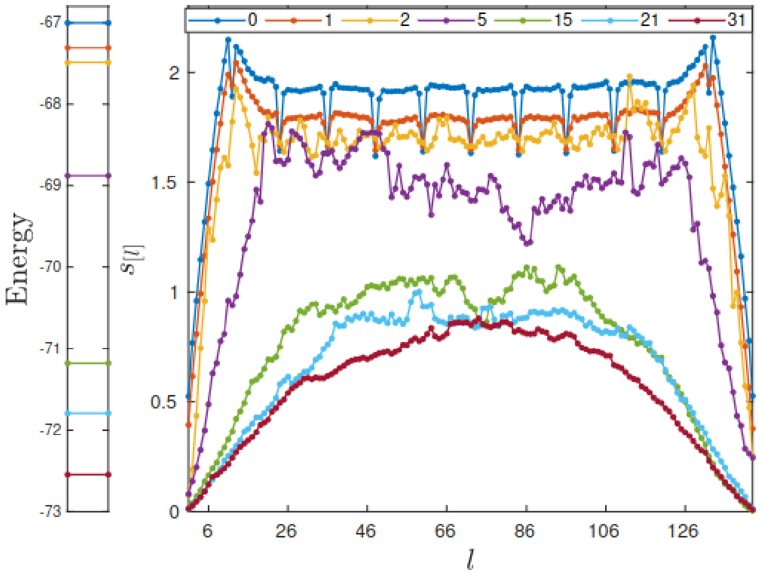


- Every unitary can be factorized as product of 2-site unitaries
- Any permutation can be reached by nearest-neighbor swaps

The unitary can be factorized, but cannot be locally optimized in this factorized form….

Idea: reorder the orbitals after several sweeps of DMRG with local optimization

Fiedler-vector based reordering



- Strong compression of block entropy
- Undamped fluctuations close to optimum
- DMRG restart is needed after reorderings

Systematic optimization with swap gates

Parametrization of unitaries close to 1: $G(U) = e^{i\sum_{i<j}\vartheta_{i,j}c_i^\dagger c_j}$

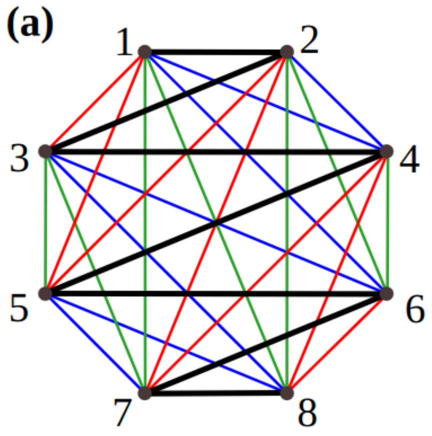
The local algorithm can only touch $\vartheta_{i,i+1}$

Idea: reorder the sites systematically to make every mode-pair nearest neighbors

Old Graph-theory problem

[22] É. Lucas, *Récréations Mathématiques*, Vol. 1 (Gauthier-Villars, 1882).

Walecki's construction (N guests, and a round table)



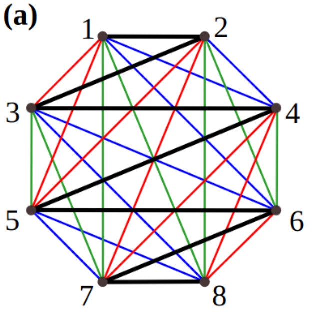
- Put nodes on a regular polygon
- First permutation: black zig-zag line
- New permutations by clock-wise rotation

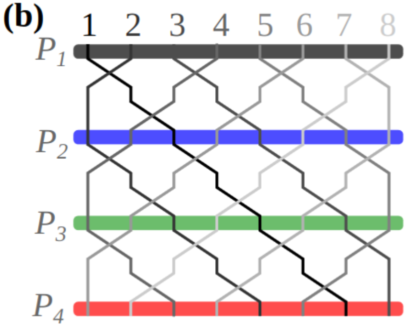
Odd number of modes?
Add one extra dummy mode and
erase it from the true permutations

Systematic optimization with swap gates

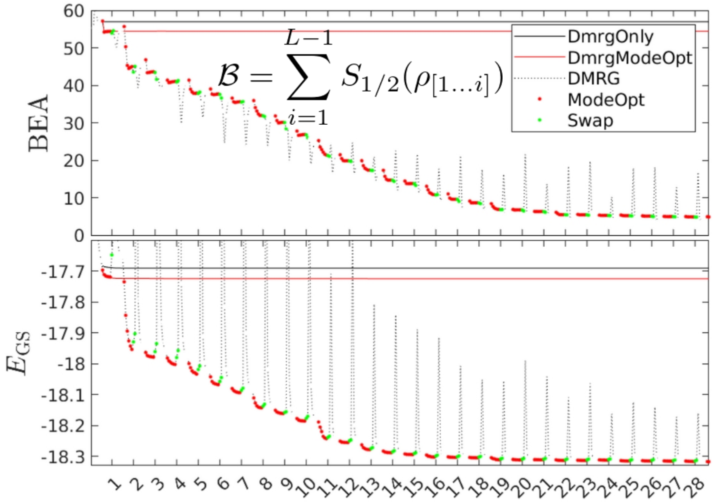
Nice feature of Walecki's construction:

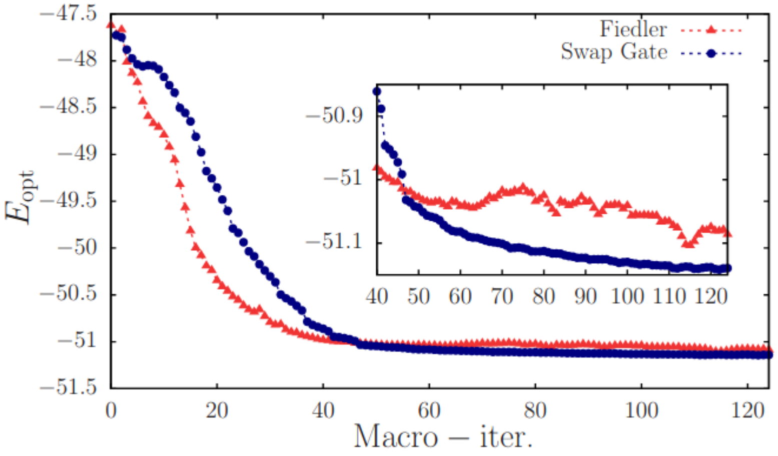
Consecutive permutations can be easily generated by 2-layers of swap gates





- Update of MPS can be done by TEBD!
- No DMRG restart is necessary
- Systematic approach





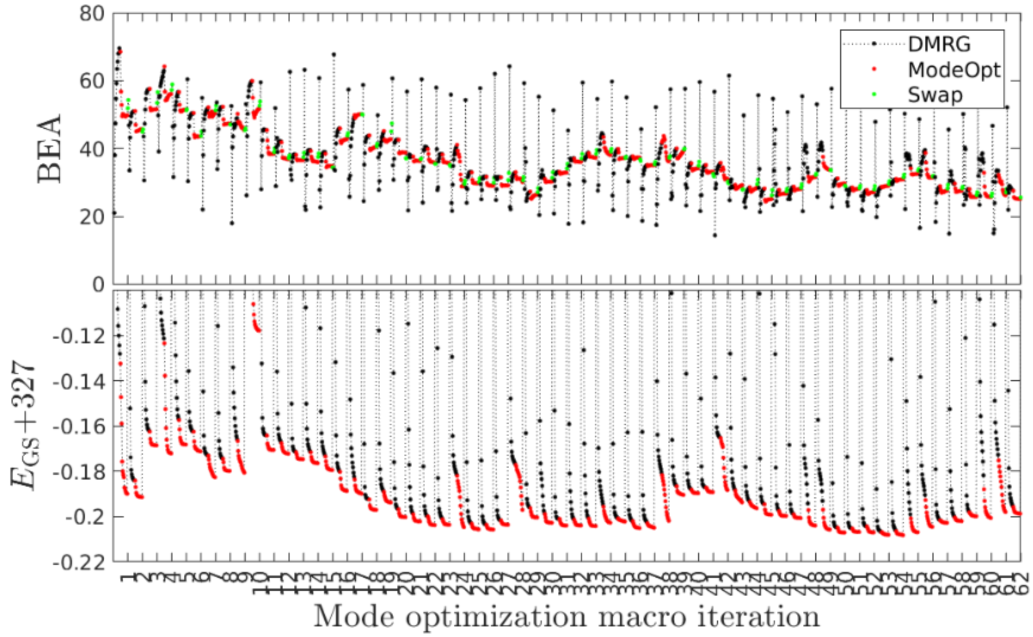
Some issue with the swap-gate based approach

Cost function $$\mathcal{B} = \sum_{i=1}^{L-1} S_{1/2}(\rho_{[1\ldots i]})$$

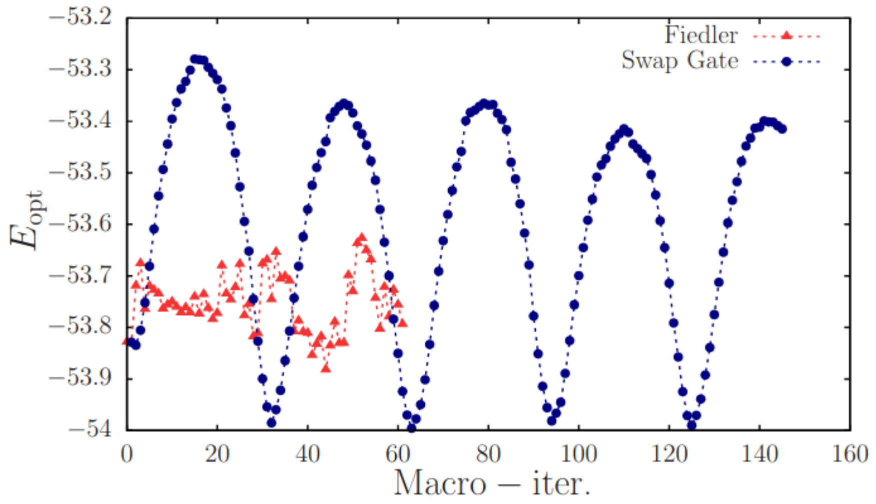
This is defined for a given permutation

Reordering the orbitals may increase the cost!

Next sweeps of local MO may not recover it



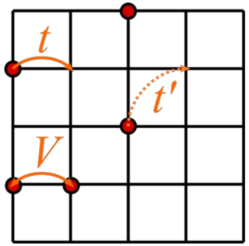
Molecular system Fe_4S_4

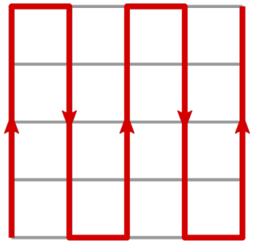


Repulsive 2D Hubbard

2D local lattice models: locality vs. entanglement optimization

Local basis:





- Locality: $O(N^{(d-1)/d})$ terms in the effective Hamiltonian
- Some phases (e.g. Mott inslator) are well captured by DMRG.
- Difficulties at metallic phases and critical points.
- PBC: higher numerical cost

$$\mathcal{H} = \sum_{\langle i,j \rangle} \left(t\, c_i^\dagger c_j + h.c \right) + \sum_{\langle i,j \rangle'} \left(t'\, c_i^\dagger c_j + h.c \right) + \sum_{\langle i,j \rangle} V n_i n_j$$

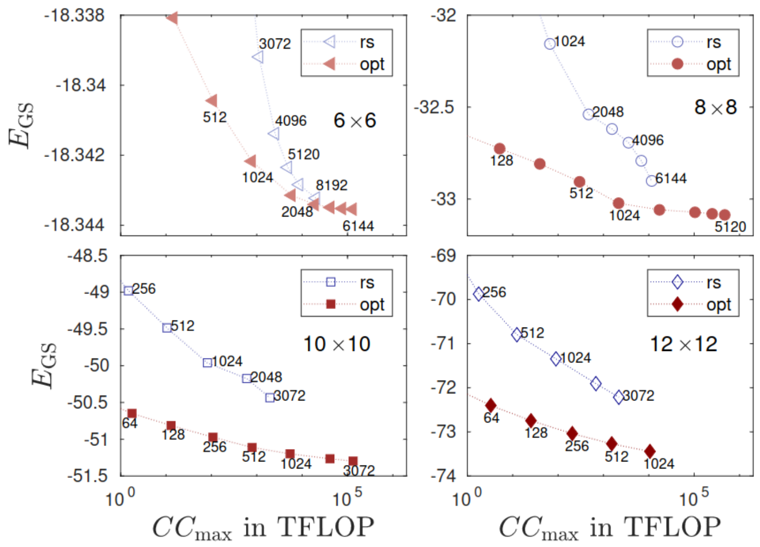
Optimized basis:

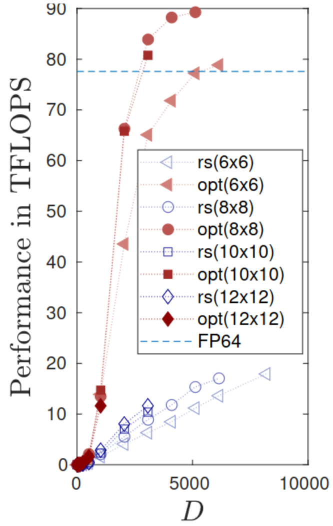
$$H = \sum_{ij} T_{ij} c_i^\dagger c_j + \sum_{ijkl} V_{ijkl} c_i^\dagger c_j^\dagger c_k c_l$$

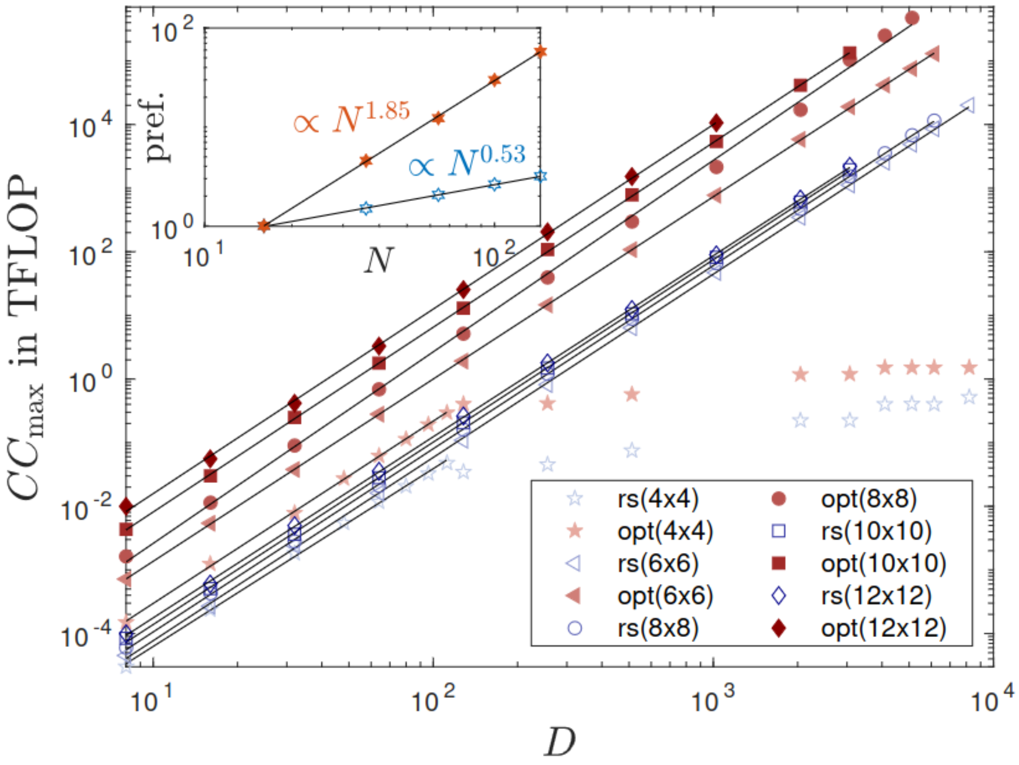
- Local interaction becomes long-ranged
- N^4 terms in the Hamiltonian
- MPO bond dimension ~ N^2 instead of $N^{0.5}$
- Bond dimension probably much lower

Question: which basis is more efficient numerically?

Local vs optimal basis: numerical complexity?







- The numerical complexity (# of floating point operation) is 2-4 orders of magnitude smaller in the optimized basis at the same precision!
- GPU parallelization is much more efficient for dense Hamiltonian

Open Questions

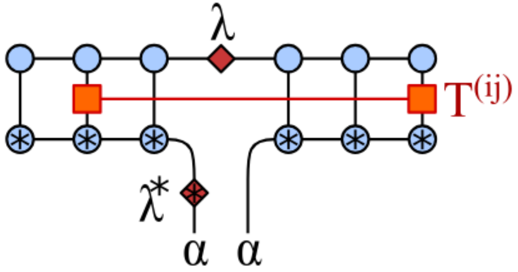
$$\mathcal{B} = \sum_{i=1}^{L-1} S_{1/2}(\rho_{[1\dots i]})$$

Idea: gradient descent method?

$$\frac{\partial \mathcal{B}}{\partial \vartheta_{i,j}}$$

Can be expressed through the derivative of Schmidt weights

$$\frac{\partial p_\alpha}{\partial \vartheta_{i,j}}$$



Non-interacting (long-ranged hopping) Hamiltonian, that compresses entanglement

Solve time-evolution with BUG, stop when BEA is minimal

Time evolution with mode optimization?

- Efficient implementation of long-ranged Hamiltonian: we are not forced to local basis!
- Entanglement barrier is probably much lower if optimal basis is used.
- Lower entanglement → longer simulation times.

Mode optimization beyond the active space

Questions:

- Is the HF-based active space selection good? → **CAS-SCF** ?
- If it is good, is the single particle basis optimal for DMRG? → **Entanglement based mode optimization**
- What is the optimal ordering of orbitals for MPS truncation? → **Mutual info. based optimal ordering**

HF-based active space selection:

Solve the HF equations
Orbitals close to the Fermi level are active
Other orbitals are frozen (totally empty or full)

↔

This HF-level selection
is probably sub-optimal
for strongly correlated systems

CAS-SCF basic Idea:

HF-based active space selection: $$H = \sum_{ij} T_{ij} c_i^\dagger c_j + \sum_{ijkl} V_{ijkl} c_i^\dagger c_j^\dagger c_k c_l$$

Solve for the ground state: $|\Psi\rangle$

Keep $|\Psi\rangle$ fixed, and rotate the orbitals (both active and frozen ones)

$$E(U) = \sum_{ij} T(U)_{ij} \langle c_i^\dagger c_j \rangle + \sum_{ijkl} V(U)_{ijkl} \langle c_i^\dagger c_j^\dagger c_k c_l \rangle$$

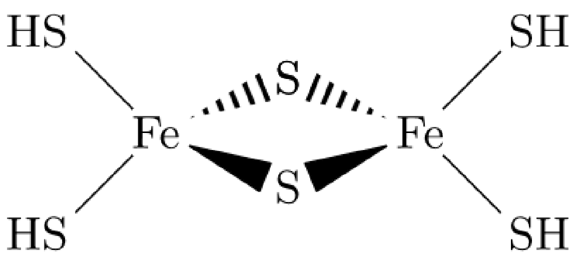
Rotated integrals

1- and 2-particle
“reduced density matrices”

Make one step towards the negative gradient, re-calculate integrals and perform a next round of Full-CI or DMRG

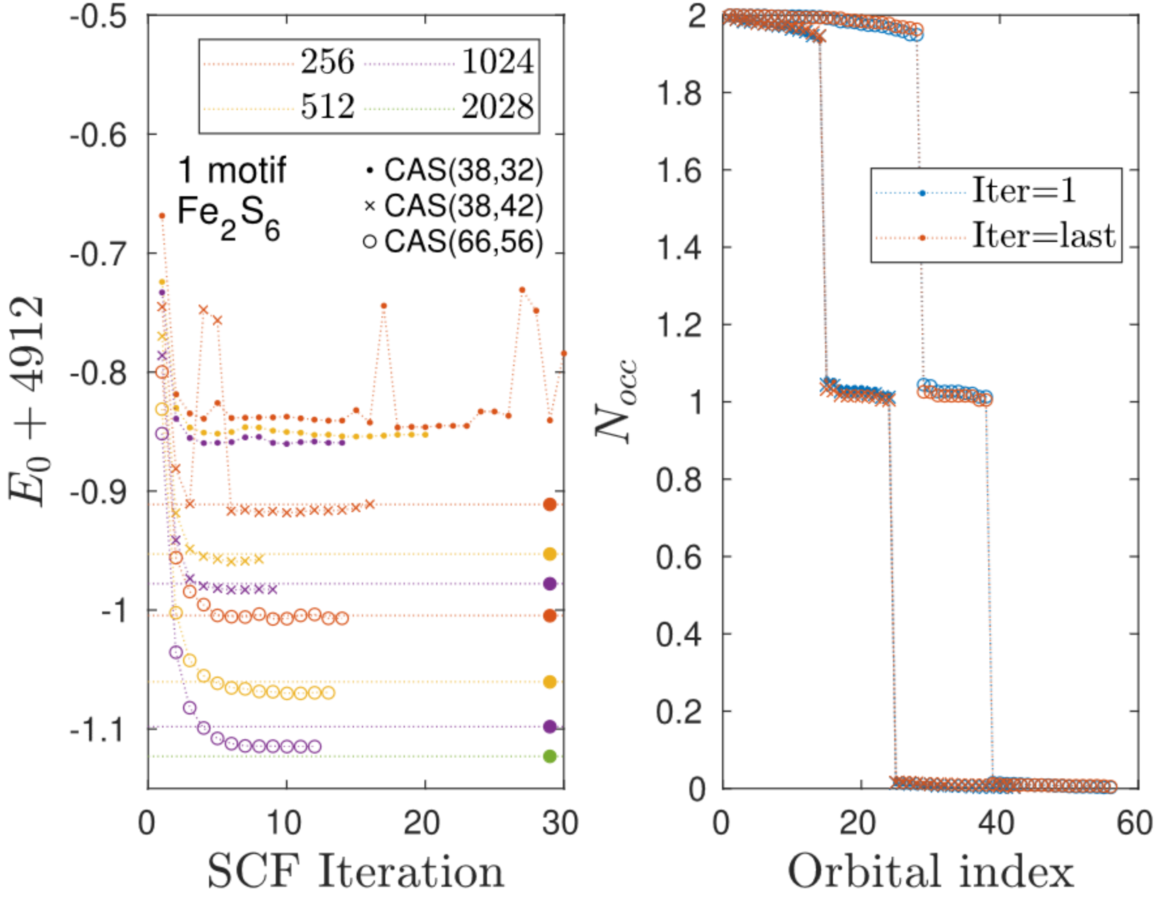
DMRG-SCF results

$Fe_2S_6H_4$



Two iron atoms: 5/2 spin

10 orbitals with unpaired occupation



Summary

Questions:

- Is the HF-based active space selection good? → **DMRG-SCF**
- If it is good, is the single particle basis optimal for DMRG? → **Entanglement based mode optimization**
- What is the optimal ordering of orbitals for MPS truncation? → **Mutual info. based optimal ordering**

EXAM INFORMATION:

* Project deadline: 1 day before the selected exam. Send the source code in e-mail
  At the exam: present the solution
  (highlight the most important parts of source code, and show and discuss results)
* For those who make a 'literature reading' exam: write an e-mail to me to clarify
  which topic you have chosen.
* Old-school type exam: 6 topics will be sent this week.

Topics: 1.) ED, 2.) Block-RG, NRG, infinite-DMRG,
3.) finite chain DMRG and Matrix Product Operators,
4.) Matrix Product States 5.) Real Time simulations (TEBD, BUG, TDVP)
6.) Lindblad and Fermionic Mode Opt

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